

Lab 13 Activity

Part 1: Logistic Regression

1

Load in the `MplsStops` data from the `carData` package.

```
data(MplsStops, package = "carData")
summary(MplsStops)
```

Look at the documentation to understand the context. Based on this information alone, select 2 or 3 predictors that you think might predict whether a citation was issued.

2

Not all rows have information about whether a citation was issued. Remove these rows. Then, use `mutate` to transform `citationIssued` into a numeric variable of 0's (no citation) and 1's (yes citation).

3

Try out a series of models using the variables that you identified in (1). For learning's sake, you may first remove rows with missing values on the predictors, but you might also try to figure out how missing data methods might apply to logistic regression. Use classification accuracy and model comparisons to select the best-performing model. What do you find?

Part 2: Multilevel Regression

For this part, we'll work with a dataset that contains weight measurements on multiple occasions for Asian children born in Britain. Load in the necessary packages and the dataset using the following code:

```
library(tidyverse)
library(haven)
library(lme4)
dat <- read_dta("Lab_13_asian.dta")
dat %>% glimpse #already in long format
```

Rows: 198

Columns: 6

```
$ id      <dbl> 45, 45, 45, 45, 45, 258, 258, 258, 258, 287, 287, 287, 287, 483~
$ occ     <dbl> 1, 2, 3, 4, 5, 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 1, 2, 1, 2, 1, ~
$ age     <dbl> 0.1368925, 0.6570842, 1.2183436, 1.4291581, 2.2724161, 0.191649~
$ weight  <dbl> 5.171, 10.860, 13.150, 13.200, 15.880, 5.300, 9.740, 9.980, 11.~
$ brthwt  <dbl> 4140, 4140, 4140, 4140, 4140, 3155, 3155, 3155, 3155, 3850, 385~
$ gender  <dbl> 1, 1, 1, 1, 1, 2, 2, 2, 2, 1, 1, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1, ~
```

- id: child identifier
- occ: numbered occasion of each measurement
- age: child's age (in years) at each measurement
- weight: child's measured weight (kg)
- brthwt: child's birthweight
- gender: 1 = male, 2 = female

4

Fit a simple (no covariates) variance components model predicting child's weight using these data. What problem do you run into? Considering the nature of these data, what do you think might be the problem?

5

Add one covariate to the model that you think will lead to the highest residual ICC. Calculate the residual ICC.

6

Given the variables available in this dataset, what combination of variables do you think might give the most realistic and accurate model of weight? Be creative, and remember what you learned about regression - many of the same principles apply! What ICC do you get?